

A Miniaturized Low-SAR Antenna for Breast Tumor Diagnosis

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Abstract— This research introduces a novel compact wideband patch antenna ($25 \times 25 \times 0.79$ mm³) on a Rogers RT5880 substrate for biomedical tumor detection. Operating across 6.68–8.82 GHz with a stable radiation pattern of 4 dBi gain SAR below 1.37 W/Kg. Tumor detection was verified in breast phantom via CST simulations, where shifts in resonant frequency, reflection coefficients (S_{11}) and variations in electric and magnetic field intensities indicated malignancy, maintaining compliance with international SAR guidelines. This design addresses challenges of size, penetration depth, and SAR in existing models, making it a promising candidate for non-invasive biomedical imaging and diagnostic applications.

Keywords— Breast tumor, WBAN, wideband antenna, low SAR, reflection co-efficient, biomedical applications, cancer detection, phantom.

I. INTRODUCTION

Wireless health monitoring systems increasingly integrate advanced antennas to ensure efficient communication with low power consumption and compliance with SAR limits (<1.6 W/kg for 1g tissue) [1]. Miniaturized and flexible designs improve wearability while enabling continuous monitoring through electromagnetic reflections. Diagnostic applications, particularly breast cancer detection, have gained importance, as conventional methods such as mammography and MRI are costly and expose patients to radiation. Breast cancer affects 2.3 million women annually worldwide, with early detection improving survival by up to 90%. Despite lower incidence rates, Black women face a 41% higher mortality rate than White women [2]. Anyone, regardless of gender, age, ethnicity, body size, or location, can be affected by a tumor [3]. Operating across 3.1–10 GHz, UWB antennas are well-suited for biomedical imaging and IEEE 802.15.6 Wireless Body Area Network (WBAN) applications. Their wide bandwidth further supports accurate tumor localization through backscattered signal analysis, enabling early detection with minimal patient risk. Various studies demonstrate their effectiveness in non-invasive diagnostics. For instance, a $341.76 \times 34 \times 0.508$ mm³ wearable monopole array operating at 3.2–4.2 GHz achieved -32.5 dB port isolation while maintaining SAR limits of 1.59 W/kg for 1 g tissue at 50 mW input power [4]. Flexible patch antennas using Kapton polyimide at 5.8 GHz showed slight resonance

shifts [5], while textile-based ISM-band antennas achieved 7.39 dBi gain, $>96\%$ efficiency, and SAR values of 1.37 W/kg [6]. An FR-4 based design ($56.9 \times 56.9 \times 1.6$ mm³) operating in 2.243–2.835 GHz demonstrated -23.7 dB return loss but reported SAR up to 11.1 W/kg in tumor presence, raising safety concerns [7]. Slotted RT Duroid 5880 patch antennas at 5.8 GHz exhibit SAR <1.6 W/kg [8] but require improved resolution in array configurations. Overall, microwave imaging leverages dielectric property contrasts permittivity and conductivity between healthy and malignant tissues, enabling effective tumor localization through antenna-based biomedical systems.

In this paper a miniaturized antenna ($25 \times 25 \times 0.79$ mm³) with novel shape addresses the challenges of large size, high Specific Absorption Rate (SAR), and inadequate penetration depth, particularly in the case of breast tumor tissue. Its wideband range (6.68–8.82 GHz) enabling high-resolution imaging and deep tissue penetration simulated with breast phantom model at different tumor position and showed enhanced performance characteristics, resolution and sensitivity also SAR levels (1.37 W/Kg) making it safer for biomedical applications. Section II describes the geometrical shape of the proposed antenna, while section III discusses its behavior in free space. The process of detecting human breast tumors is detailed in sections III and IV. Lastly, section IV provides the conclusion along with a critical evaluation of the study.

II. ANTENNA DESIGN

Ultra-Wideband (UWB) antennas are widely used in tumor detection due to their broad frequency coverage, tissue penetration capability, and improved imaging resolution. Low return loss enhances detection by minimizing energy reflection. Rectangular and circular microstrip patch antennas are the most common UWB configurations. In this work, a novel microstrip patch antenna is proposed for the 6.68–8.82 GHz band using a Rogers RT5880 substrate. The design process involved multiple optimization stages (Fig. 1). The initial S-shaped patch with a full ground plane was modified in stage-I by introducing diagonal and parasitic rectangular arms with a circular ring, producing a notch near 8 GHz (stage-II). In stage-III, replacing the full ground with a partial

ground plane height of 5 mm improved bandwidth and lowered return loss. Further refinement in stage-IV by introducing ground edge cuts enhancing gain (4 dB), bandwidth and achieve -40.73 dB return loss. The optimized S-shaped parasitic patch with a partial ground plane effectively reduces substrate energy absorption, lowers the quality factor, and broadens bandwidth, making it highly suitable for microwave-based tumor imaging.

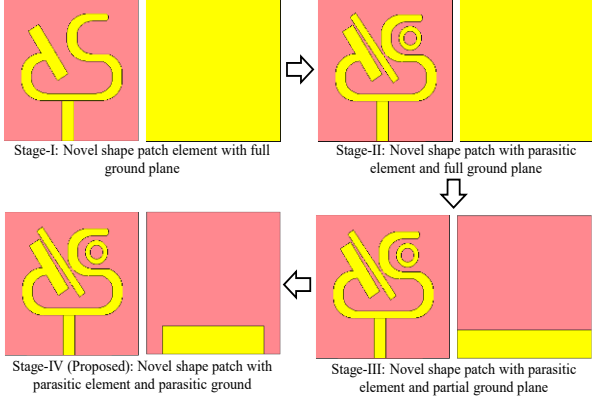


Fig. 1. Design evolution of proposed antenna.

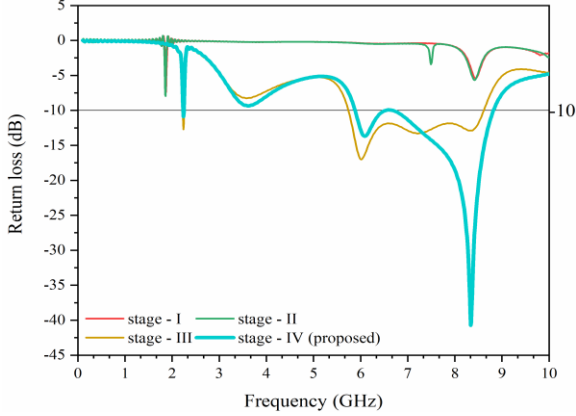


Fig. 2. Variation of reflection coefficient across all antenna design stages.

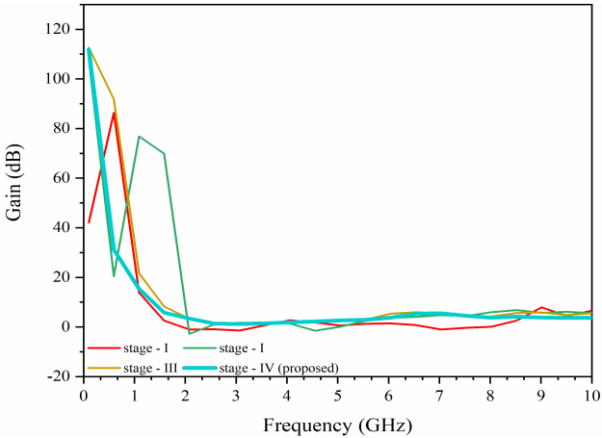


Fig. 3. Gain response at successive antenna design stages.

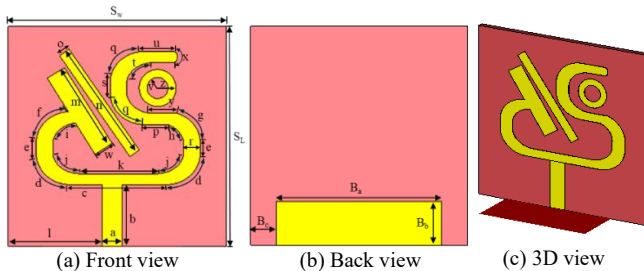


Fig. 4. Proposed novel shape patch antenna.

TABLE I. PARAMETRIC VALUES OF PROPOSED NOVEL SHAPED ANTENNA

Parameter	Dimension (mm)	Parameter	Dimension (mm)	Parameter	Dimension (mm)
a	2.20	k	8.74	u	3.27
b	7.00	l	10.80	v	1.20
c	10.92	m	10.18	w	2.28
d	5.25	n	14.00	x	1.67
e	1.29	o	1.30	y	2.73
f	5.06	p	2.66	z	2.10
g	4.17	q	5.25	B _a	19.00
h	2.90	r	1.88	B _b	5.00
i	3.54	s	1.29	B _c	3.00
j	3.97	t	3.97	S _w = S _L	25.00

Fig. 2 and 3 depict the progression of reflection coefficient and gain throughout the different stages of the antenna design process. To obtain the S_{11} parameter within the desired range, we focus on fundamental equations (1-4) [9]. Below is the mathematical expression for calculating the dimensions of the rectangular antenna:

Calculation of the width (W):

$$width = \frac{c}{2f_0 \sqrt{\frac{\epsilon_R + 1}{2}}} \dots \dots \dots (1)$$

Where, f_0 = Resonance frequency (Hz), ϵ_R = Relative permittivity of the dielectric substrate (F/m), $c = 3 \times 10^8$ (m/s).

Calculation of the effective dielectric constant (ϵ_{eff}) is related with the height (h), dielectric constant (ϵ_R) and width (w) of the patch antenna and expressed as:

$$\epsilon_{eff} = \frac{\epsilon_R + 1}{2} + \frac{\epsilon_R - 1}{2} \left[\frac{1}{\sqrt{1 + 12 \left(\frac{h}{w} \right)}} \right] \dots \dots \dots (2)$$

Calculation of effective length L_{eff} :

$$L_{eff} = \frac{c}{2f_0 \sqrt{\frac{\epsilon_R + 1}{2}}} \dots \dots \dots (3)$$

Calculation of actual length of the patch:

$$Length = \frac{c}{2f_0 \sqrt{\epsilon_{eff}}} - 0.824 \left[\frac{(\epsilon_{eff} + 0.3) \left(\frac{w}{h} + 0.264 \right)}{(\epsilon_{eff} - 0.258) \left(\frac{w}{h} + 0.8 \right)} \right] \dots \dots \dots (4)$$

Initially, the dimensions of the antenna radiator are determined using the patch antenna equation, which takes into account the resonant frequency and relative dielectric permittivity (ϵ_r). Ground measurements reveal dimensions of $19 \text{ mm} \times 5 \text{ mm}$, while the substrate dimensions are $S_L = 25 \text{ mm}$ (length) and $S_W = 25 \text{ mm}$ (width), obtained through parametric studies using simulation software. The ground plane of the antenna significantly influences bandwidth improvement. A 50Ω microstrip feedline, 2.3 mm wide and 7 mm long, connects to the patch. The antenna is mounted on an economically efficient Rogers RT5880 substrate with a dielectric constant of 2.3, dielectric tangent-loss of 0.0009, and a thickness of 0.79 mm. Additionally, a rectangular parasitic element measuring $14 \times 1.30 \text{ mm}^2$ and a 0.9 mm wide circular ring are attached to the radiating patch. The proposed novel patch antenna is illustrated in Fig. 4, showing the front, back, and 3D views in parts (a), (b), and (c), respectively. Detailed antenna parameters are summarized in Table I. The design and simulation were conducted using the microwave-based simulator CST Microwave Studio.

III. ANALYSIS OF ANTENNA RESULTS AND PERFORMANCE

A. Antenna performance in free space:

The optimized antenna achieves a return loss of -40.73 dB at 8.34 GHz with a wide impedance bandwidth of 6.68–

8.82 GHz, primarily due to ground-plane modifications and parasitic element integration that improve impedance matching plotted in Fig. 5. Radiation patterns (Fig. 6) confirm stable directional behavior, with E- and H-plane lobes oriented at 97° and 90° , yielding magnitudes of 15.9 dB (V/m) and -33 dB (A/m). The 3D far-field distribution (Fig. 7) shows a 3D directivity of 4.483 dBi, indicating efficient energy focusing. Gain and efficiency analysis with respect to frequency in Fig. 8, demonstrate 4 dB gain (peak 5.5 dB at 7.03 GHz) with 90.42% efficiency, attributed to reduced substrate absorption, ensuring reliable biomedical performance.

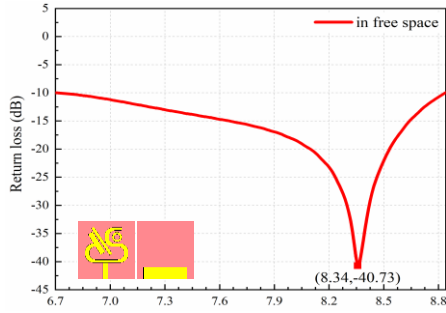


Fig. 5. Reflection co-efficient of the proposed antenna.

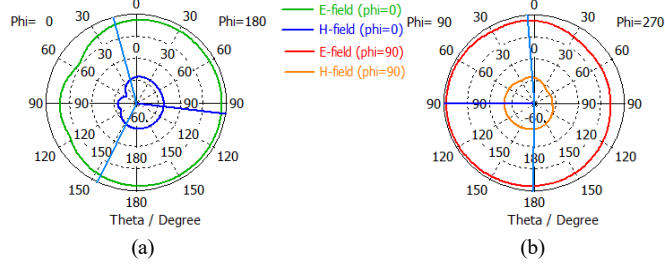


Fig. 6. Simulated radiation patterns in the polar coordinates at (a) $\phi=0^\circ$ and (b) $\phi=90^\circ$.

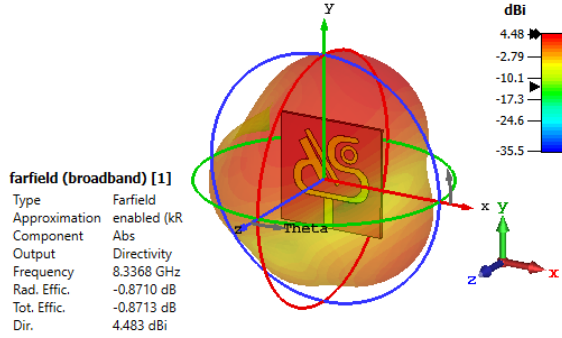


Fig. 7. Simulated directivity of proposed antenna in free space (3D configuration).

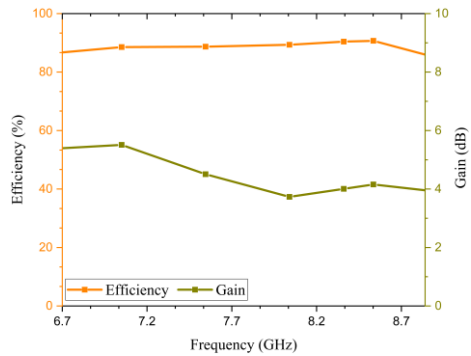


Fig. 8. Efficiency and gain analysis of proposed antenna in free space.

B. Antenna performance with healthy human breast model:

A breast phantom model with four dielectric layers, (properties in Table II) was simulated in CST Studio Suite to evaluate microwave penetration. The antenna, positioned 20 mm from the model, exhibited a resonance shift to 8.315 GHz with return loss -37.29 dB (Fig. 10a). Gain and directivity were -5.65 dB and 5.88 dBi, respectively in Fig. 10(b), 11. Radiation patterns in Fig. 11 show main lobes at 98° ($\phi=0^\circ$) with beamwidth of 44.2° . Side lobes remain minimal. SAR analysis in Fig. 12, confirmed safety, yielding 0.867 W/Kg (1 g) and 0.659 W/Kg (10 g), within FCC (Federal Communications Commission) and ICNIRP (International Commission on Non-Ionizing Radiation Protection) limits.

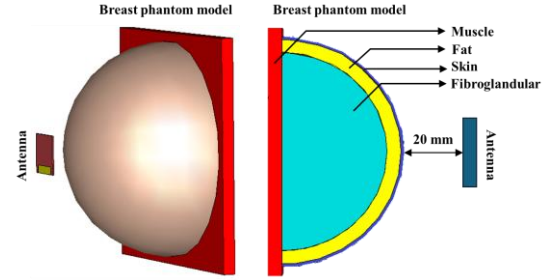


Fig. 9. Antenna setup with breast phantom model (3D view and cross-sectional view)

TABLE II: BREAST TISSUES PROPERTIES

Tissue	Permittivity	Conductivity (S/m)	Density (Kg/m ³)	Thermal conductivity (W/K/m)	Thickness (mm)	Diameter (mm)
Skin	39.4	1.1	1010	0.29	$t=2$	140
Fat	10	0.1	911	0.62	$t=8$	136
Muscle	50	1.91	1040	0.38	$t=8$	150
Glandular	12.5	0.31	1035	0.56	—	120
Tumor	67	5	1058	0.56	—	10

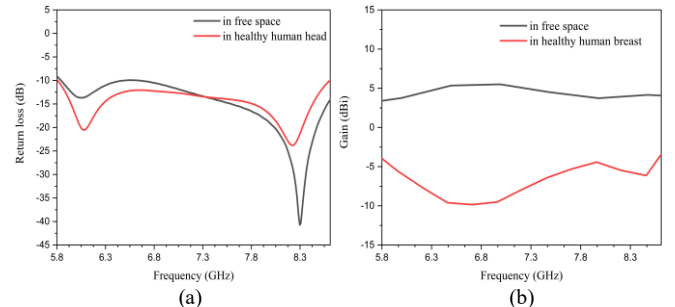


Fig. 10. Simulated comparison between free space and healthy human breast in terms of (a) return loss (S_{11}) and (b) gain.

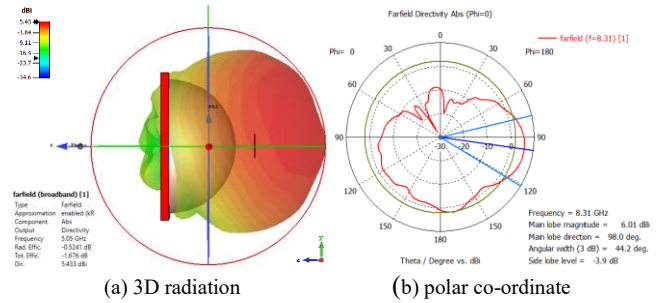


Fig. 11. Directivity analysis between healthy human breast and free space.

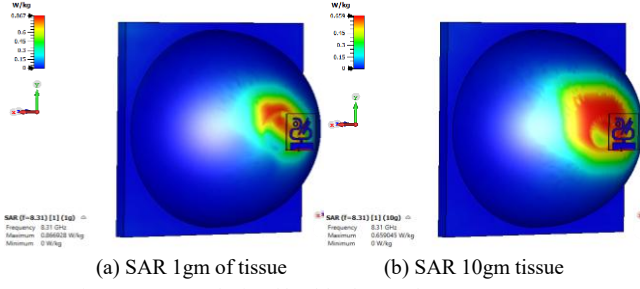


Fig. 12. SAR analysis of healthy human breast at 8.31 GHz.

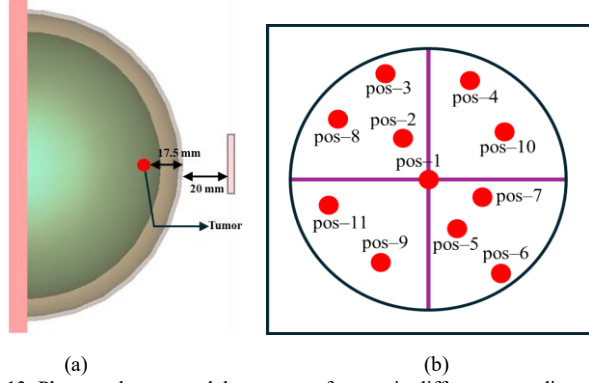


Fig. 13. Phantom breast model presence of tumor in different co-ordinate.

C. Antenna performance with human breast model induced with tumor:

For breast tumor detection, a spherical tumor of 5 mm radius was modeled at a depth of 17.5 mm inside glandular tissue (Fig. 13a). The tumor was placed at 11 different positions (Fig. 13b) to evaluate the effect of location on detection accuracy, with results summarized in Table III. Simulation results in Fig. 14 confirm the capability of antenna to distinguish tumor presence. As shown in Fig. 14, the introduction of a tumor shifts the resonance frequency due to modified dielectric properties of surrounding tissue. At position (10, 12, 37.5), return loss rises to -51.801 dB at 8.31 GHz, confirming detectability. Tumor tissue, with higher water content and permittivity, interacts differently with microwaves, producing variations in reflection coefficient amplitude (Table IV). Field distribution analysis (Fig. 15) reveals reduced electric field intensity from 22063.5 to 21107.9 V/m and magnetic field intensity from 64.32 to 62.27 A/m with a tumor present. This variation arises from absorption, reflection, and scattering within tumor tissue, which also alters signal propagation and antenna gain. Directivity patterns in Fig. 16 further differentiate tissue states. Whether Fig. 17(a) Specific Absorption Rate (SAR) analysis highlights localized absorption in tumors at 8.31 GHz, SAR values were 0.8669 W/kg for healthy tissue and 1.0035 W/kg for tumor tissue (1 g mass), while 10 g averaged SAR was 0.6664 W/kg (Fig. 17b). Overall, the antenna

effectively differentiates healthy and malignant tissues within 5.78–8.71 GHz based on return loss, VSWR, E/H-field variations, gain, and SAR (Table IV).

TABLE III OUTCOMES BASED ON DIFFERENT POSITIONS OF THE TUMOR

Position	Resonance (GHz)	Return loss (dB)	SAR (1gm)	Depth (mm)
10, 12, 37.5	8.31	-52.84	1.0035	17.5
15, 30, 43	8.30	-37.632	0.9752	23
22, 50, 60	8.315	-37.829	0.8665	40
2, 60, 78	8.315	-55.582	0.8473	58
7, -10, 50	8.3	-57.709	0.8172	30
-17, -25, 70	8.315	-38.028	0.8662	50
-6, 0, 55	8.285	-39.14	0.8371	35
28, 40, 67	8.315	-38.031	0.8639	47
15, -28, 59	8.305	-37.605	0.8644	39
-29, 22, 53	8.325	-66.806	0.8328	33
44, -17, 56	8.315	-37.976	0.8654	36

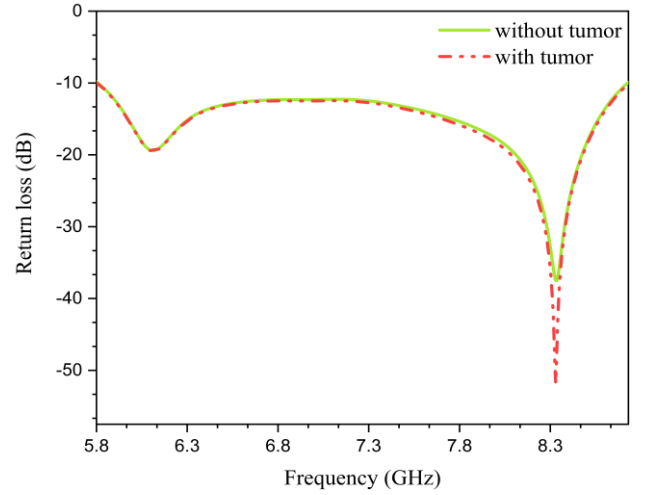


Fig. 14. Return Loss (S_{11}) curve analysis between with and without presence of tumor.

TABLE IV: COMPARISON OF PARAMETER FOR BREAST PHANTOM

Parameter	In free space	Without tumor	With tumor
Resonance (GHz)	8.34	8.315	8.31
SAR for 1g (W/Kg)	—	0.8669	1.0035
SAR for 10g (W/Kg)	—	0.659	0.6664
Electric field intensity (V/m)	84223.7	22063.5	21107.9
Return loss (dB)	-40.73	-37.289	-51.801
VSWR	1.02	1.027	1.0084
Efficiency (%)	90.42	81.47	81.5
Gain (dBi)	4	-5.65	-5.38
Directivity (dB)	4.75	5.88	5.93

IV. COMPARATIVE ANALYSIS

Its miniaturized structure overcomes limitations such as inadequate gain and bandwidth found in existing models. The comparison in Table V highlights its superior performance compared to contemporary antennas in [4–8] and [10–12], making it ideal for biomedical imaging applications that demand simplicity, compactness, and seamless integration.

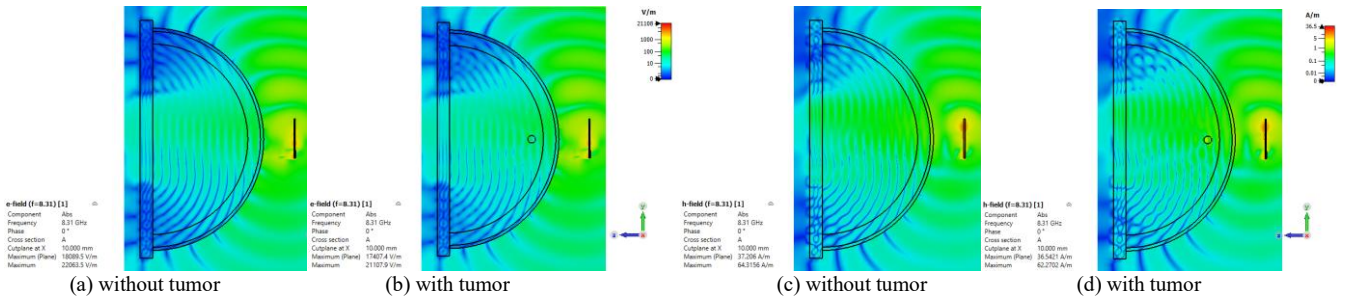


Fig. 15. E-field (a-b) and H-field (c-d) distribution along with and without presence of tumor in breast phantom.

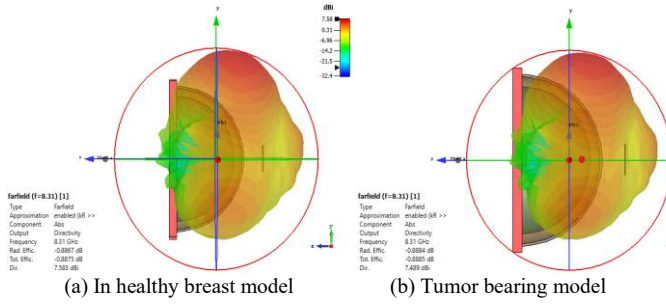


Fig. 16. Directivity at 8.31 GHz in absence and presence of tumor in breast phantom model.

V. CONCLUSION AND DISCUSSION

A compact wideband patch antenna, designed for biomedical applications, is presented in this paper. The prototype, constructed from Rogers RT5880 substrate, measures $25 \times 25 \times 0.79$ mm³ and operates within a 6.68–8.82 GHz bandwidth. Miniaturization reduces antenna size while preserving performance, enabling its use in compact biomedical and wearable systems for breast tumor detection. Simulation models of human phantoms demonstrated their tumor detection efficacy through blue shifts in resonant frequency and reflection coefficients from 7.79 GHz to 7.87

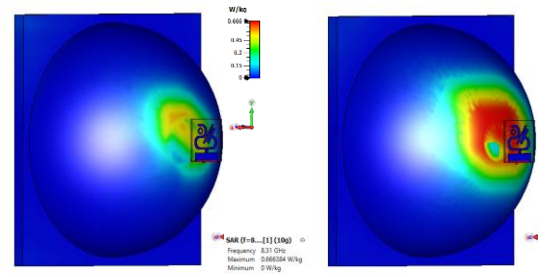


Fig. 17. Effect of SAR in presence of tumor inside human breast phantom model.

GHz and from -13.66 dB to -15.98 dB respectively. Additionally, the presence of a tumor caused notable changes in the electric field (E-field) and magnetic field (H-field) deflections, from 22063.5 to 21107.9 V/m. The analysis, summarized in Table IV, evaluates SAR, VSWR, directivity, and radiation patterns in both tumor-present and tumor-absent scenarios, concludes that this single antenna is proficient in identifying anomalies within breast model. Furthermore, the design adheres to international safety standards set by the ICNIRP and FCC, ensuring its reliability and safety for biomedical imaging and diagnostic applications.

TABLE V. COMPARATIVE STUDY OF THE PROPOSED ANTENNA WITH RECENT RELEVANT WORK

Ref. no.	Size (L×W×h) mm ³	Substrate material	Operating frequency range (GHz)	Reflection coefficient (dB)	Centre freq. (GHz)	Gain (dBi)	Efficiency (%)	Max SAR for 1g (W/kg)	Distance between antenna to phantom (mm)
4	341.76×34×0.508	Rogers RT5880	3.2–4.2	≈-22	3.7	—	—	1.59	50
7	56.9×56.9×1.6	FR4	2.243–2.835	-30.53	2.4	6.35	—	11.1462	1
5	26.68×34.84×1.6	Kapton Polyimide	5.7–5.9	-20.832	5.8	5.203	—	1.27	115
6	50×70×1	Textile (Jeans)	5.73–5.85	-24.054	5.79	7.39	96	1.37	—
10	90×40×1.6	FR4	2.26–2.47	-40.26	2.38	4.63	—	0.179–0.083	10–50
11	59×76×1.5	FR4	2.2–2.6	-19	2.63	—	—	—	On surface
8	51×35×0.508	RT Duroid 5880	5.725–5.875	-29.9	5.8	—	26.26	1.1	3–9
12	37.58×29.24×1.6	FR4	2.46–2.48	-25.446	2.46	—	—	0.04906	On surface
Prop. Work	25×25×0.79	Rogers RT5880	6.68–8.82	-40.73	8.34	4	90.42	1.37	20

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